

DiagnoCenter

Diagnostic Center Billing ERP

Synology NAS Docker Installation Guide • Version 2026.05

For Synology DSM 7.x with Container Manager

What You Need

- **Synology NAS** with DSM 7.x (DSM 6.x also works with the older Docker package)
- **Container Manager** installed from Package Center
- **SSH enabled** (needed only for the first setup)
- **At least 4 GB RAM** recommended (2 GB minimum)
- **1 GB free disk space** for the app + space for your data
- **Any computer with a web browser** to access the ERP — no installation needed

Why use a NAS? Your Synology NAS runs 24/7 anyway. The ERP server stays online even when all clinic computers are turned off. Plus, NAS backups are easy and reliable.

Model recommendation: Any x86-based Synology (DS220+, DS420+, DS920+, DS1522+, etc.) works well. ARM-based models (like DS120j) will run slower but still work.

Step-by-Step Installation

1 Install Container Manager

1. Open **Package Center** on your Synology
2. Search for **Container Manager**
3. Click **Install**
4. Wait for installation to complete (may take a few minutes)

Screenshot: Package Center showing Container Manager installed

2 Enable SSH (One-Time Setup)

You need SSH only for copying files. You can disable it later.

1. Open **Control Panel** → **Terminal & SNMP**
2. Check **Enable SSH service**
3. Set port to **22** (or your preferred port)
4. Click **Apply**

Screenshot: Control Panel → Terminal & SNMP settings

3 Create a Shared Folder

The app files and database will live in a shared folder on your NAS.

1. Open **File Station** on your Synology
2. Click **Create** → **Create New Shared Folder**
3. Name it **diagno-center** (or any name you prefer)
4. Click **OK**

4 Upload Project Files

Copy the project folder into your new shared folder. You can do this via:

- **File Station web interface:** Drag and drop the files
- **Network share (SMB):** Map the NAS drive on your computer and copy
- **SFTP/SCP:** If you are comfortable with command line

The folder should look like this when done:

```
diagno-center/  
  docker-synology/  
    docker-compose.yml  
    .env.example  
    README.md  
  docs/  
  postgres_data/ ← created automatically later
```

5 Configure the Environment

Tell the app your settings (password, port, etc.).

1. In File Station, go to `diagno-center/docker-synology/`
2. Find the file `.env.example`
3. Right-click → **Rename** → change to `.env`
4. Double-click to edit (or download, edit, upload back)

Change at minimum these two lines:

```
# Change this to a strong password
DB_PASSWORD=changeme

# Change this only if port 8888 is already in use
HOST_PORT=8888
```

Important: The default password is `changeme`. You MUST change this before going live. Patient data lives in this database.

6 Build and Start (via SSH)

Now use SSH to tell Docker to build and start everything.

1. On your computer, open a terminal or command prompt
2. SSH into your Synology (replace with your NAS IP):

```
ssh admin@192.168.1.100
```

Enter your admin password when asked.

3. Navigate to the project folder:

```
cd /volume1/diagno-center/docker-synology
```

4. Build and start:

```
docker compose up -d --build
```

Note: The first build takes 5-10 minutes because it downloads Node.js and builds the app. Be patient. On slow NAS models this may take up to 20 minutes.

7 Create Database Tables

Run the database migration once to create all tables:

```
docker compose run --rm migrate
```

You will see a lot of output ending with something like:

```
Using config 'drizzle.config.ts'
[drizzle-kit] pushing...
Done!
```

Tip: You only need to run this once during installation. If you ever update the app to a new version, run it again to apply schema changes.

8 Open the App

On any computer in your clinic:

1. Open **Google Chrome** or **Microsoft Edge**
2. In the address bar, type your NAS IP with the port:

```
http://192.168.1.100:8888
```

(Replace `192.168.1.100` with your actual NAS IP address)

3. Press Enter
4. The login screen appears — log in with your staff PIN

Find your NAS IP: On the Synology, go to **Control Panel** → **Network** → **Network Interface**. Look for "IPv4 Address."

Daily Use

Task	How to Do It
Check if running	Container Manager → Container tab → look for <code>diagno-web</code> and <code>diagno-api</code> with green status
Restart all services	Container Manager → select all 3 containers → Action → Restart
View logs	Container Manager → select container → Logs tab
Stop everything	SSH in and run: <code>docker compose down</code>
Start everything	SSH in and run: <code>docker compose up -d</code>

Updating the App

When a new version is available:

1. Copy the new project files over the old ones (keep your `.env` file)
2. SSH into the NAS:

```
cd /volume1/diagno-center/docker-synology
docker compose down
docker compose up -d --build
docker compose run --rm migrate
```

3. Your data in `postgres_data/` is preserved — nothing is lost

Backing Up (Critical)

Method 1: Copy the Database Folder (Fastest)

1. Open **File Station**
2. Navigate to `diagno-center/docker-synology/postgres_data/`
3. Right-click → **Copy to** → choose another shared folder or USB drive

Do this weekly at minimum. The database folder contains ALL patient data.

Method 2: Container Manager Export

1. Container Manager → **Project** → select `diagno-center`
2. Click **Action** → **Export**
3. This creates a `.dcp` file with the whole stack

Method 3: Automated Daily Backup

Set up Synology Hyper Backup to back up the `postgres_data` folder daily to another location (another NAS, USB drive, or cloud).

Your database is your clinic's lifeblood. Without regular backups, a NAS hard drive failure could permanently destroy all patient records, bills, and reports. Back up weekly at minimum. Daily is better.

Network Access

Local Network (Default)

By default, the app is only available inside your clinic's network. Anyone on the same Wi-Fi can open `http://nas-ip:8888`.

Access from Outside (Optional)

If you need to access the ERP from home or another clinic:

Option A: VPN (Recommended, Most Secure)

1. Install **VPN Server** from Package Center
2. Set up OpenVPN or L2TP/IPSec
3. Connect your device to the VPN
4. Open the app as if you were in the clinic

Option B: Reverse Proxy with HTTPS

1. Get a domain name (e.g., from Namecheap or GoDaddy)
2. In DSM: **Control Panel** → **Login Portal** → **Advanced** → **Reverse Proxy**
3. Set up a reverse proxy rule pointing to `localhost:8888`
4. Enable HTTPS with Let's Encrypt

Option C: QuickConnect (Not Recommended)

Synology QuickConnect works but is slower and gives less control. Not recommended for medical data.

Troubleshooting

Problem	Solution
"Cannot GET /" in browser	Wait 2 more minutes. The API container may still be starting. Check Container Manager → Logs.
Port 8888 already in use	Edit <code>.env</code> and change <code>HOST_PORT=8888</code> to another number like <code>HOST_PORT=8889</code> . Then restart with <code>docker compose up -d</code> .
Container won't start	Check RAM usage: Control Panel → Resource Monitor . Need at least 2 GB free. Close unused packages or add more RAM.
"Database connection error" in logs	The <code>diagno-db</code> container must be running and "healthy" before <code>diagno-api</code> starts. Restart everything: <code>docker compose restart</code> .
Very slow performance	Your NAS may be ARM-based (slower CPU). Consider upgrading to an x86 model (DS220+, DS420+, etc.).
Build takes forever	First build downloads Node.js and compiles the app. This is normal. Future updates will be much faster.
Lost the NAS IP address	Go to Control Panel → Network → Network Interface → look for "IPv4 Address."

Hardware Recommendations

Clinic Size	RAM	Storage	Example NAS Model
1-5 staff	4 GB	50 GB free	Synology DS220+
5-15 staff	8 GB	100 GB free	Synology DS420+
15+ staff	16 GB	200 GB free	Synology DS920+ or DS1522+

Security Checklist

- ☐ Change the default `DB_PASSWORD` in `.env` before going live
- ☐ Keep DSM and Container Manager updated via Package Center
- ☐ Enable DSM firewall and restrict port 8888 to your clinic's IP range
- ☐ Set up regular backups (Hyper Backup or manual copy)
- ☐ Use HTTPS if exposing to the internet (reverse proxy with Let's Encrypt)
- ☐ Disable SSH after setup if you don't need it regularly
- ☐ Change default admin PIN after first login

Need Help?

Contact your software support and provide:

1. Your NAS model and DSM version (found in **Control Panel** → **Info Center**)
2. Screenshots of any error messages from Container Manager → Logs
3. Which step you got stuck on